Link-Level Simulation of a 5G NR OFDM Downlink

Link Budget, Waveform Generation, and BER Performance under AWGN, Rician, and Rayleigh Fading (n78 Band, 100 MHz)

Ghada Mokhtar

Telecommunications Engineering Student — ENSTA (École Nationale Supérieure des Technologies Avancées)

MATLAB R2024b — 5G Toolbox & Communications Toolbox

Abstract

This report presents a complete link-level study of a simplified 5G NR downlink operating in the n78 band (3.5 GHz, 100 MHz channel bandwidth, 30 kHz subcarrier spacing). The work spans three stages: (1) a manual link budget calculation based on the 3GPP TR 38.901 Urban Macro (UMa) propagation model, producing reference SNR operating points for line-of-sight (LOS) and non-line-of-sight (NLOS) conditions; (2) an OFDM waveform simulation, first built manually from first principles (IFFT/FFT, cyclic prefix, 16-QAM) and later reimplemented using MATLAB's 5G Toolbox (nrCarrierConfig, nrPDSCH, nrOFDMModulate); and (3) bit error rate (BER) performance evaluation under AWGN, Rician (LOS), and Rayleigh (NLOS) fading channels. Three implementation errors were identified and resolved during the verification process, each described here with its root cause and fix, as this diagnostic process is itself a central part of the engineering value of the exercise.

1. Introduction and Objective

The objective of this project is to build, from first principles, a verified link-level simulation of a 5G NR downlink connection, connecting a manual RF link budget calculation to an OFDM waveform simulation and a bit-error-rate performance analysis. The scenario models a single user served by an urban macro (UMa) base station at 3.5 GHz (band n78), under both line-of-sight and non-line-of-sight propagation conditions.

The project deliberately avoids relying on black-box tools: every stage — from the Friis transmission equation through to the OFDM cyclic prefix and the QAM decision boundaries — was derived and implemented explicitly, then cross-checked numerically, so that each result can be explained and defended rather than merely reproduced.

2. System Model and Parameters

Table 1 summarizes the system-level parameters used throughout the study.

Parameter	Value
Frequency band	n78 (3.5 GHz)
Channel bandwidth	100 MHz
Subcarrier spacing (SCS)	30 kHz
Resource blocks (5G Toolbox model)	273 RB (per 3GPP TS 38.101-1, Table 5.3.2-1)
Modulation	16-QAM, Gray-coded
Channel coding	None (uncoded link, focus on modulation + channel effects)
Spatial layers	1 (no MIMO)
Propagation model	3GPP TR 38.901 Urban Macro (UMa)
BS antenna height / UE height	25 m / 1.5 m

Table 1 — System-level simulation parameters.

3. Link Budget Analysis

The link budget was derived manually, starting from the Friis free-space path loss equation and extending to the 3GPP UMa empirical path loss model, for a base station-to-user distance of 500 m (horizontal), with a 3D distance of 500.55 m after accounting for antenna height difference.

3.1 Free-Space Path Loss (reference)

$$L_FSPL\ (dB) = 20 \cdot log10(d[km]) + 20 \cdot log10(f[MHz]) + 32.44$$

For d = 0.5 km and f = 3500 MHz: L_FSPL ≈ 97.3 dB.

3.2 3GPP UMa Path Loss (LOS and NLOS)

$$PL_UMa\text{-}LOS\ (dB) = 28.0 + 22 \cdot log10(d3D) + 20 \cdot log10(fc[GHz])$$

$$PL'_UMa\text{-}NLOS\ (dB) = 13.54 + 39.08 \cdot log10(d3D) + 20 \cdot log10(fc[GHz]) - 0.6(hUT - 1.5)$$

with PL_NLOS = max(PL_LOS, PL'_NLOS). Evaluating these for the scenario above gives PL_LOS ≈ 98.27 dB and PL_NLOS ≈ 129.91 dB — a gap of roughly 32 dB between the two propagation conditions, illustrating why line-of-sight planning and beamforming are critical for mid-band 5G coverage.

3.3 Full Link Budget

$$P_RX\ (dBm) = P_TX + G_TX + G_RX - L_cable - L_path - L_shadow - L_body$$

Using representative values (P_TX = 46 dBm, G_TX = 17 dBi, G_RX = 0 dBi, L_cable = 2 dB, shadow fading margin = 8 dB, body loss = 3 dB) and a receiver noise floor of −87 dBm (thermal noise + 7 dB noise figure over 100 MHz), the resulting reference points are summarized in Table 2.

Condition	Path Loss (dB)	Received Power (dBm)	Reference SNR (dB)
LOS	98.27	−48.27	≈ 38.7
NLOS	129.91	−79.91	≈ 7.0

Table 2 — Link budget results and derived SNR reference points, used as markers on the BER curves in Section 6.

4. OFDM Waveform Simulation Methodology

Two implementations were built, in sequence, reflecting an iterative development process.

4.1 Manual Implementation (first-principles baseline)

The first version constructs the OFDM waveform manually: 16-QAM symbols are mapped onto 200 active subcarriers within a 256-point FFT (the remainder reserved as guard bands), transformed to the time domain via a unitary IFFT, and a cyclic prefix (25% of the FFT length) is prepended per OFDM symbol to protect against inter-symbol interference. This version was built to be runnable under MATLAB R2014, and therefore avoids toolbox features introduced in later releases, normalizing symbol power by hand.

4.2 5G Toolbox Implementation

After upgrading to MATLAB R2024b with 5G Toolbox access, the simulation was reimplemented using standard 3GPP-compliant functions: nrCarrierConfig and nrPDSCHConfig define the carrier and physical downlink shared channel (PDSCH) allocation; nrPDSCH performs standard-compliant bit scrambling, modulation, and layer mapping; nrOFDMModulate/nrOFDMDemodulate handle the OFDM waveform generation and reception using the exact numerology tables specified by 3GPP. This version is structurally much closer to what would be used in an industrial 5G link-level simulator.

5. Channel Models and Assumptions

Three channel conditions were evaluated at each SNR point:

- AWGN only — an ideal baseline with no fading, used to validate the simulation against the closed-form theoretical BER curve.
- Rician fading (representing LOS) — a dominant direct path plus a weaker scattered component, modeled with a K-factor of 10 dB (a representative assumed value, not derived from the specific link budget scenario above).

- Rayleigh fading (representing NLOS) — no dominant path, purely scattered energy (equivalent to the Rician model with K = 0).

Two simplifying assumptions were made explicitly and are flagged here rather than left implicit:

- Flat fading: a single complex channel gain is applied per OFDM symbol (constant across all subcarriers within that symbol), rather than a fully frequency-selective multipath model. This is reasonable when the channel coherence bandwidth exceeds the signal bandwidth, but is a simplification relative to a full tapped-delay-line (e.g. 3GPP TDL) channel model.
- Perfect equalization: the receiver is assumed to know the exact fading gain and divides it out exactly (genie-aided channel knowledge), rather than estimating it from reference (DMRS) signals as a real receiver would.

6. Verification and Debugging

A deliberate numerical cross-verification process (including an independent Python re-implementation of the algorithm) was used to validate the simulation at each stage. This process surfaced three distinct implementation errors, each described below with its diagnosis and resolution.

6.1 Issue 1 — Y-axis scaling hid the true BER behavior

Symptom: the simulated BER curve appeared to plateau near 1 across most of the SNR range on the plot. Diagnosis: the theoretical BER curve decays to numerically negligible values (as low as 10^{-300}) at high SNR, causing MATLAB to auto-scale the y-axis to fit this range — compressing the realistic, measurable BER range (limited by the finite simulated bit count, $\approx 2.5\times10^{-6}$) into an unreadably small portion of the plot. Fix: the y-axis was explicitly bounded to $[10^{-6}, 1]$, matching the resolution actually achievable by the simulation.

6.2 Issue 2 — Theoretical BER formula used Es/N0 where Eb/N0 was expected

Symptom: the simulated BER curve was consistently worse than the theoretical reference curve by a roughly constant SNR offset (5-7 dB), despite the simulation logic itself being correct. Diagnosis: the standard Gray-coded M-QAM BER formula, P_b = (4/k)(1 − 1/√M)·Q(√(3k·Eb/N0/(M−1))), takes Eb/N0 (energy per bit) as its argument. The SNR value used throughout this simulation (and set via MATLAB's awgn(...,'measured') option) represents Es/N0 (energy per symbol). Substituting Es/N0 directly in place of Eb/N0 left an extra factor of k = log2(M) = 4 inside the square root, making the theoretical curve artificially optimistic. Fix: the formula was corrected to Q(√(3·Es/N0/(M−1))), which is algebraically equivalent to substituting Eb/N0 = Es/N0 / k into the original formula. This was confirmed numerically in Python, after which the simulated and theoretical curves matched closely.

6.3 Issue 3 — Missing PDSCH descrambling (5G Toolbox version)

Symptom: after switching to the 5G Toolbox implementation, the simulated BER remained flat at approximately 0.5 across the entire SNR range (0-40 dB), for all three channel conditions. Diagnosis: nrPDSCH applies 3GPP-standard pseudo-random bit scrambling internally before modulation. The function used to recover bits, nrSymbolDemodulate, reverses only the modulation step, not the scrambling. Comparing the still-scrambled recovered bits against the original transmitted bits is statistically equivalent to comparing against an independent pseudo-random sequence, yielding BER ≈ 0.5 regardless of SNR (visually close to the top of a log-scale BER plot spanning 10^{-6} to 1). Fix: nrSymbolDemodulate was replaced with nrPDSCHDecode, the correct inverse counterpart to nrPDSCH, which performs layer de-mapping, soft-bit (LLR) computation, and descrambling together. After this fix, the AWGN curve matched the theoretical reference and the Rician/Rayleigh curves showed the expected relative degradation.

7. Results

Figure 1 shows simulated BER versus SNR for the three channel conditions, alongside the theoretical AWGN reference curve, with the two link-budget-derived SNR operating points (Section 3.3) marked as vertical reference lines.

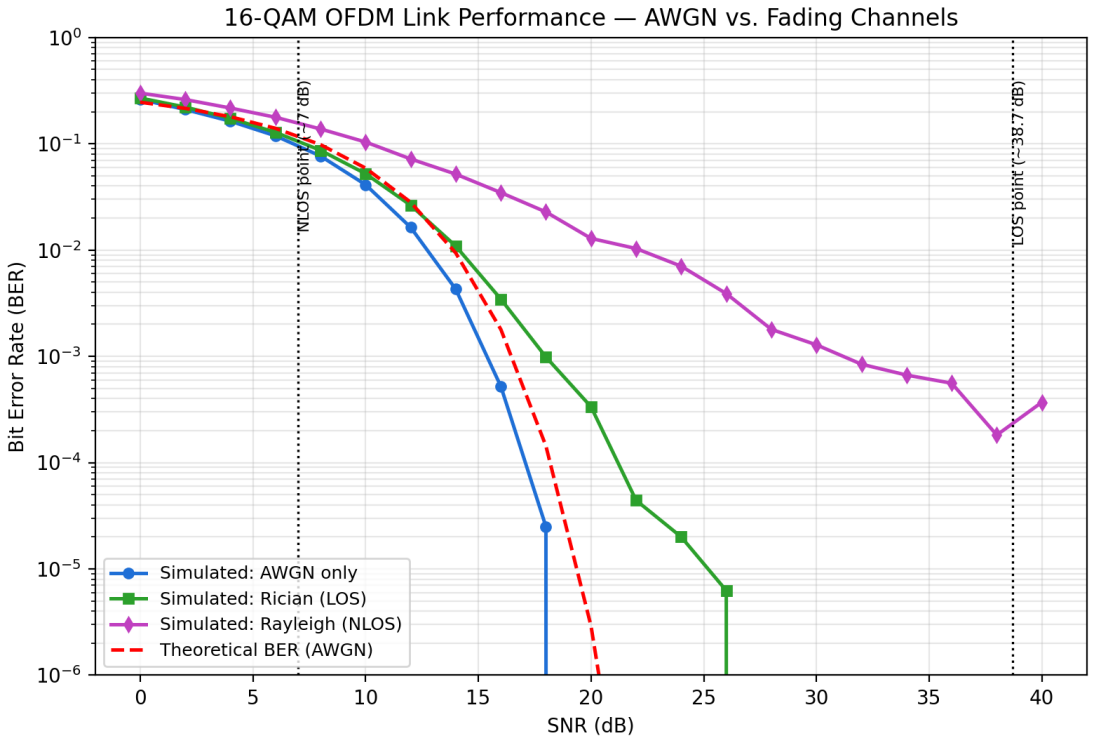


Figure 1 — Simulated BER for AWGN, Rician (LOS), and Rayleigh (NLOS) channels, vs. the theoretical AWGN reference curve. Vertical dotted lines mark the link-budget-derived NLOS (~7 dB) and LOS (~38.7 dB) SNR operating points.

Three observations follow directly from Figure 1:

- The simulated AWGN curve closely tracks the theoretical reference, confirming the correctness of the modulation, OFDM, and channel implementation after the fixes in Section 6.
- At the link-budget-derived LOS operating point (~38.7 dB), both the AWGN and Rician curves are already well into the error floor — the link operates with negligible bit errors, consistent with the very large link margin computed in Section 3.3.
- At the NLOS operating point (~7 dB), BER is on the order of 10^{-1} to 10^{-2}, reflecting a usable but considerably degraded link — consistent with the ~32 dB path loss penalty derived in Section 3.2, and illustrating quantitatively why LOS conditions and techniques such as beamforming matter for mid-band 5G deployments.

8. Conclusion and Future Work

This project connects a manual RF link budget to a validated, standards-based OFDM link-level simulation, and demonstrates a disciplined verification process that identified and corrected three distinct implementation errors. The resulting simulation reproduces expected physical behavior: close agreement with theory under AWGN, graceful degradation under Rician fading, and the well-known slower BER decay characteristic of Rayleigh fading without diversity.

Natural extensions, not attempted in this version, include: replacing the flat-fading assumption with a full 3GPP tapped-delay-line channel model (e.g. TDL-C for NLOS, TDL-D for LOS); replacing perfect channel knowledge with DMRS-based channel estimation; and adding LDPC channel coding to evaluate coded, rather than uncoded, link performance.

References

3GPP TR 38.901, "Study on channel model for frequencies from 0.5 to 100 GHz."

3GPP TS 38.101-1, "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone."

J. G. Proakis and M. Salehi, Digital Communications, 5th ed., McGraw-Hill.

MathWorks 5G Toolbox Documentation — nrCarrierConfig, nrPDSCHConfig, nrPDSCH, nrPDSCHDecode, nrOFDMModulate.